

Question	Working	Answer	Mark	Notes
21		$\frac{2}{5}$	P1	for process to find $\overrightarrow{AB} (= \mathbf{b} - \mathbf{a})$ or $\overrightarrow{BA} (= \mathbf{a} - \mathbf{b})$
			P1	for process to find $\overrightarrow{MN} (= -\frac{1}{2}\mathbf{b} + \mathbf{a} + 2\mathbf{a})$ or $\overrightarrow{PN} (= -k(\mathbf{b} - \mathbf{a}) + 2\mathbf{a})$ or $\overrightarrow{MP} (= -\frac{1}{2}\mathbf{b} + \mathbf{a} + k(\mathbf{b} - \mathbf{a})$ or $\frac{1}{2}\mathbf{b} + (1 - k)(\mathbf{a} - \mathbf{b}))$
			P1	for process to find two of $\overrightarrow{MN}$, $\overrightarrow{PN}$ and $\overrightarrow{MP}$
			P1	for process to find k , using $\overrightarrow{MN}$ as a multiple of $\overrightarrow{PN}$ or using $\overrightarrow{MN}$ as a multiple of $\overrightarrow{MP}$ or using $\overrightarrow{PN}$ as a multiple of $\overrightarrow{MP}$
			A1	for $\frac{2}{5}$ oe